

Imperfection for Precision: Upcycling Imperfect Data for High-Precision Robotic Manipulation

Hao Wei^{1*}, Yang Liu^{1*}, Chao Tang^{1*}, Shengbao Li¹, Jiangtao Chen¹, Jinxuan Zhu^{1,2}, Jiaheng Wang¹,
Hong Yin¹, Zhaofeng Cao^{1,3}, and Tingguang Li^{1†}

¹Samsung Robotics eXperience ²National University of Singapore ³University of Illinois Urbana-Champaign

Abstract—Training vision-language-action (VLA) models for high-precision manipulation typically requires task-specific, high-quality data (e.g., teleoperation), which is slow and expensive to collect. To reduce this burden without compromising manipulation precision, we propose $\epsilon 4P$ (Imperfection for Precision), a simple yet effective method that “upcycles” two otherwise discarded data sources: (1) low-precision data from the target task and (2) high-precision data from mismatched tasks. Rather than naively mixing these imperfect data sources throughout co-training, $\epsilon 4P$ controls where each source contributes along the flow-matching trajectory. Specifically, low-precision, target-task data is used at high noise to preserve high-level task context and high-precision, task-mismatched data is used at low noise to transfer low-level action precision. Through real-robot experiments on both sub-millimeter, high-precision tasks and coarse-grained tasks, we demonstrate that the proposed method (1) effectively leverages additional imperfect data to improve policy performance by up to 31.7 percentage points, and (2) can replace an equal amount of task-specific, high-quality data with an average performance drop of only 4.2 percentage points. Overall, $\epsilon 4P$ points toward a scalable paradigm for high-precision manipulation, in which heterogeneous, imperfect data can be systematically repurposed to reduce reliance on costly task-specific, high-quality data. More details are available at <https://varepsilon4p.github.io/>.

I. INTRODUCTION

Vision-language-action (VLA) models offer a promising path toward general-purpose robot manipulation by scaling on diverse robot experience [1]–[3]. However, training these models for high-precision manipulation tasks, such as insertion and assembly [4], still requires task-specific, high-quality data (i.e., teleoperation), as the tight geometric tolerances of these tasks demand highly accurate and consistent demonstrations. Collecting such data requires substantial human effort and hardware resources [5]–[9]. Meanwhile, large amounts of imperfect data may already be available, yet their quality often falls short of the stringent requirements of high-precision tasks, making them difficult to reuse directly [10], [11]. We therefore ask: **How can imperfect data be effectively upcycled for high-precision manipulation?**

Existing methods primarily address this problem through native co-training, using sampling ratios or mixture weights to control how much each data source contributes while applying its contribution uniformly across the training objective [3], [6], [12]–[15]. While effective for less demanding tasks, such uniform integration of imperfect data can

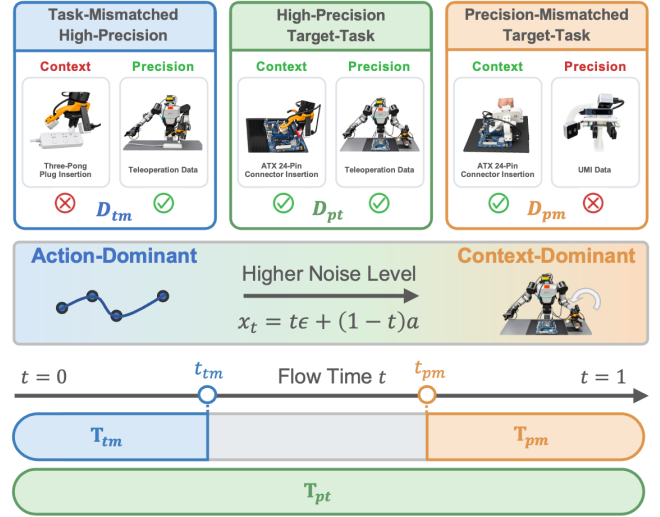


Fig. 1. **Conceptual overview of $\epsilon 4P$.** The proposed method upcycles two types of imperfect data sources by routing them to source-dependent flow time admission regions for high-precision manipulation.

introduce conflicting supervision and bias the policy away from the precise behaviors, a limitation also observed in prior work [16]. Collecting substantially more task-specific, high-quality data could in principle mitigate this issue, but doing so is costly and difficult to scale. Conversely, simply increasing the amount of imperfect data does not necessarily resolve the mismatch and may even amplify conflicting supervision. This creates a practical dilemma: the most reliable data are difficult to scale, while the most scalable data cannot be naively integrated.

In this work, rather than treating imperfect data as “trash”, we argue that the key is to determine where along the flow-matching trajectory each data source can provide useful supervision. Based on this insight, we propose $\epsilon 4P$ (Imperfection for Precision), which routes imperfect data to source-dependent admission regions along the flow-matching trajectory of a policy [17] (e.g., a VLA), as illustrated in Figure 1. We consider two complementary sources of imperfect data: (1) low-precision target-task data collected with UMI [9] and (2) high-precision teleoperation data from mismatched tasks. To determine the admission region for each source, $\epsilon 4P$ uses two complementary mechanisms tailored to their respective imperfections. For the former, we identify the flow-time region where its noised actions become sufficiently indistinguishable from high-precision demonstrations of the

*Equal contribution.

†Corresponding author.

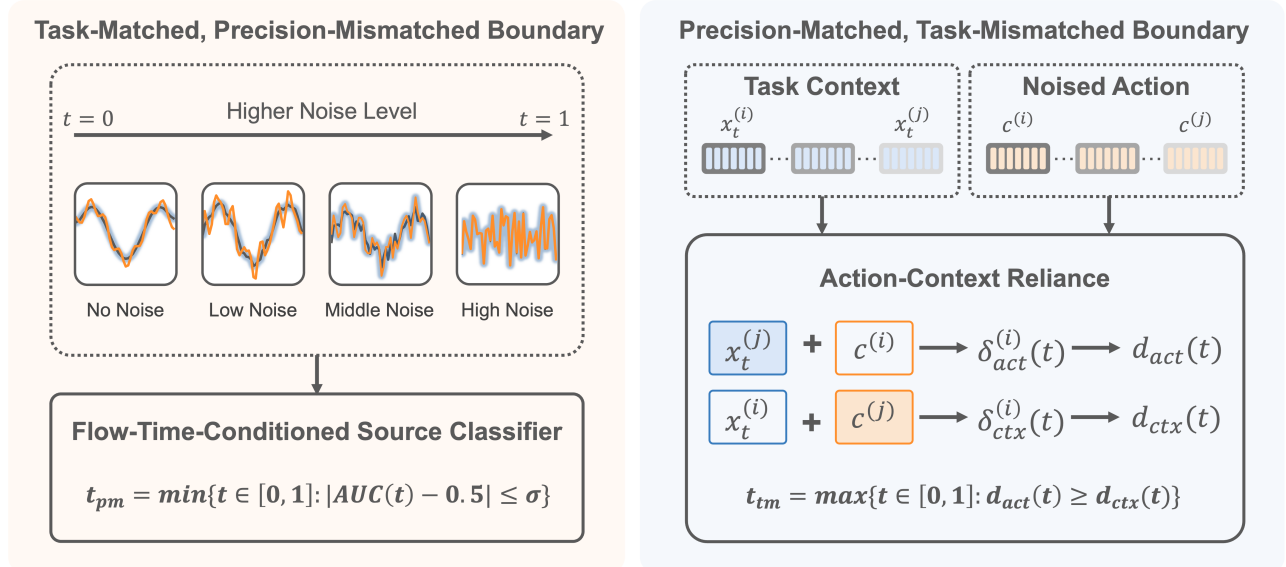


Fig. 2. An overview of two mechanisms for deriving the source-dependent admission boundaries t_{pm} and t_{tm} , tailored to task-matched, precision-mismatched data $\mathcal{D}_{pm}$ (left) and task-mismatched, precision-matched data $\mathcal{D}_{tm}$ (right), respectively.

same task. For the latter, we identify the transition from context-dominant prediction at high noise to action-dominant prediction at low noise along the flow-time axis. By restricting each source to an appropriate flow-time region, $\epsilon 4P$ effectively integrates heterogeneous imperfect data into a single policy while limiting their imperfections, without modifying the model architecture or flow-matching objective.

To demonstrate the effectiveness of $\epsilon 4P$, we conduct real-robot experiments on both sub-millimeter, high-precision tasks (two-stage cable plugging and ATX 24-pin connector insertion) and coarse-grained manipulation tasks (Bolt-Nut Sorting). The results show that $\epsilon 4P$ (1) effectively leverages additional imperfect data to improve policy performance by up to 31.7 percentage points, and (2) can replace an equal amount of task-specific, high-quality data with an average performance drop of only 4.2 percentage points. Moreover, $\epsilon 4P$ consistently outperforms native co-training and source reweighting strategies commonly used in prior work.

Contribution. We propose a novel method that upcycles heterogeneous imperfect data by routing them to source-dependent admission regions along the flow-matching trajectory, reducing reliance on costly task-specific, high-quality data and providing a scalable paradigm for high-precision manipulation.

II. RELATED WORK

A. Heterogeneous Data Co-Training

Generalist robot policies increasingly leverage heterogeneous training data from diverse tasks, domains, and data sources [6], [18]–[21]. Existing methods improve such data mixtures mainly through (1) reweighting at the domain level [12] and (2) filtering or selecting useful demonstrations at the sample level [22], [23]. For co-training with imperfect data, prior work has further explored demonstration reweighting, data or trajectory-segment selection, and training under systematic domain shifts [14], [15], [24]–[29]. These approaches show that imperfect data can

still provide useful supervision when its contribution is appropriately controlled. However, they primarily regulate how much each domain, trajectory, or sample contributes, while applying its contribution uniformly across the training objective. Such uniform integration of imperfect data can introduce conflicting supervision [30] and bias the policy away from the precise behaviors required for high-precision manipulation. In contrast, $\epsilon 4P$ determines where along the flow-matching trajectory each imperfect data source should contribute, routing it to source-dependent admission regions where it can provide useful supervision.

B. Noise and Flow-Time Scheduling

Diffusion and flow-matching models have been widely applied to generative modeling, where noise level provides a natural control axis over the generation process [17], [31]. In diffusion-based image restoration, this structure has been exploited for super-resolution, deblurring, and inpainting [32], [33], while forward diffusion can progressively attenuate unknown degradations before recovering the underlying image content [34]. More broadly, noise-dependent objective weighting shows that different diffusion/flow-time regions need not contribute equally during training [35], [36]. Related work further shows that the utility of lower-quality or biased data can vary across diffusion times [37]. Together, these works suggest that the value of information and the influence of corruption vary systematically along the diffusion or flow-matching trajectory.

In robotics, diffusion models have been widely adopted in sequential generation and robot policy learning [38], and recent work has begun to exploit non-uniform noise schedules for action generation. Diffusion Forcing introduces per-token diffusion schedules for sequence modeling [39], while Streaming Diffusion Policy applies variable noise levels across the action horizon [40]. TMRL further modulates diffusion timesteps to control conditioning strength during policy adaptation [41]. These approaches demonstrate that

different elements of a sequence can benefit from different noise levels, extending noise-dependent control from image generation to sequential decision making. Concurrent work, Ambient Diffusion Policy (ADP) [16], similarly exploits diffusion time to selectively leverage suboptimal data. Taking a step forward, $\epsilon 4P$ provides a complete mechanism to automatically derive source-dependent admission regions for precision and task mismatch.

III. APPROACH

A. Problem Formulation

We consider training a flow-matching policy (e.g., a VLA) using high-precision target-task data $\mathcal{D}_{\text{pt}}$ together with two forms of imperfect supervision: (1) precision-mismatched target-task data $\mathcal{D}_{\text{pm}}$, and (2) task-mismatched high-precision data $\mathcal{D}_{\text{tm}}$. In our setup, these two sources are instantiated using UMI and teleoperation data, respectively.

Let $s \in \{\text{pt}, \text{pm}, \text{tm}\}$ denote a data source, π the source sampling distribution, and $t \in [0, 1]$ the flow time. We associate each source with an admission region $\mathcal{T}_s \subseteq [0, 1]$. For $\mathcal{D}_{\text{pt}}$, we fix $\mathcal{T}_{\text{pt}} = [0, 1]$. Given $s \sim \pi$, the flow time is sampled uniformly from the corresponding region, $t \sim \mathcal{U}(\mathcal{T}_s)$. We refer to the unmodified setting in which $\mathcal{T}_s = [0, 1]$ for all sources as *native co-training*. This setting retains the original π and applies neither source-dependent admission nor loss reweighting. Our objective is to determine $\mathcal{T}_{\text{pm}}$ and $\mathcal{T}_{\text{tm}}$ such that each imperfect source contributes only where it provides useful supervision.

In what follows, we first examine how the policy’s reliance on the noised action input and task context varies across flow time, providing a lens into how precision and task mismatch are expressed along the flow trajectory. We then derive source-dependent criteria for $\mathcal{T}_{\text{pm}}$ and $\mathcal{T}_{\text{tm}}$.

B. How Flow Time Shifts Action-Context Reliance

$\epsilon 4P$ builds on the observation that the policy’s reliance on the noised action input and task context varies across flow time. We characterize this behavior by conducting an empirical analysis on $\pi_{0.5}$ [3], while the underlying observation is expected to extend to other flow-matching policies. Let $a \in \mathbb{R}^{H \times d}$ denote an action chunk and c the task context, including visual observations, language instruction, and proprioception. We adopt the convention that $t = 0$ corresponds to the clean action and $t = 1$ to pure noise. Given $\epsilon \sim \mathcal{N}(0, I)$, the flow path is:

$$x_t = t\epsilon + (1 - t)a \quad (1)$$

The predicted action is therefore denoted as:

$$\hat{a}(x_t, c, t) = x_t - t\hat{v}(x_t, c, t) \quad (2)$$

where $\hat{v}$ is the predicted flow velocity. We use $\hat{a}$ as an action-space readout to measure the policy’s reliance on the noised action input x_t and task context c across flow time, quantified by the change in $\hat{a}$ under input perturbations.

Specifically, we measure this input reliance using a Two-Stage Cable Plugging dataset with 100 episodes. For each sample i , we draw a reference sample j from the dataset and

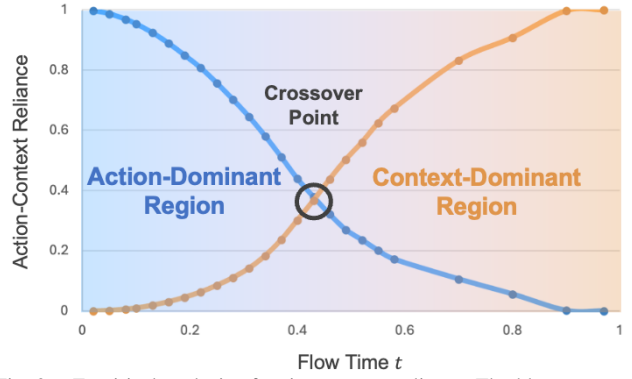


Fig. 3. Empirical analysis of action–context reliance. The blue curve represents action reliance, while the orange curve represents context reliance. For this setup, the two curves intersect at $t^* = 0.4327$.

substitute either $x_t^{(i)}$ or $c^{(i)}$ with its counterpart from j . This keeps the substituted component marginally in-distribution while measuring the policy’s sensitivity to each input. The resulting change in $\hat{a}$ can be measured as:

$$\delta_{\text{act}}^{(i)}(t) = \left\| \hat{a}(x_t^{(i)}, c^{(i)}, t) - \hat{a}(x_t^{(j)}, c^{(i)}, t) \right\|_2^2 \quad (3)$$

$$\delta_{\text{ctx}}^{(i)}(t) = \left\| \hat{a}(x_t^{(i)}, c^{(i)}, t) - \hat{a}(x_t^{(i)}, c^{(j)}, t) \right\|_2^2 \quad (4)$$

Here, δ_{act} measures the sensitivity to the noised action input, whereas δ_{ctx} measures the sensitivity to task context. These two metrics are then averaged over all samples and normalized by the sensitivity measured when both inputs are perturbed, yielding $d_{\text{act}}(t)$ and $d_{\text{ctx}}(t)$ as the final reliance scores on x_t and c at flow time t , respectively.

Figure 3 shows a single crossover from action-dominant prediction at low flow times to context-dominant prediction at high flow times. Therefore, at low flow times, $\mathcal{D}_{\text{tm}}$ can provide precise action supervision with limited influence from task mismatch; at high flow times, the target-aligned task context in $\mathcal{D}_{\text{pm}}$ becomes more beneficial despite its lower action precision.

C. Offline Admission Boundaries

The observed reliance transition motivates admitting $\mathcal{D}_{\text{tm}}$ at low flow times and $\mathcal{D}_{\text{pm}}$ at high flow times. However, since the two sources differ from $\mathcal{D}_{\text{pt}}$ along different axes, their boundaries require different diagnostics: action separability for $\mathcal{D}_{\text{pm}}$ and policy action-context reliance for $\mathcal{D}_{\text{tm}}$. Therefore, we define:

$$\mathcal{T}_{\text{tm}} = [0, t_{\text{tm}}], \quad \mathcal{T}_{\text{pm}} = [t_{\text{pm}}, 1] \quad (5)$$

where t_{tm} and t_{pm} are the admission boundaries.

Task-Matched, Precision-Mismatched Boundary. Since $\mathcal{D}_{\text{pm}}$ and $\mathcal{D}_{\text{pt}}$ demonstrate the same task, the separability of their noised actions primarily reflects differences in action precision. We therefore train a flow-time-conditioned source classifier to predict whether x_t originates from $\mathcal{D}_{\text{pm}}$ or $\mathcal{D}_{\text{pt}}$. Here, x_t is constructed by perturbing the corresponding clean action with a flow-time-dependent noise level following Equation 1, as illustrated in Figure 2 (left).

Following classifier-based two-sample testing [42] and source-separability diagnostics [43], we use its held-out AUC

to define:

$$t_{\text{pm}} = \min \{t \in [0, 1] : |\text{AUC}(t) - 0.5| \leq \sigma\} \quad (6)$$

where $\text{AUC} = 0.5$ indicates that the classifier cannot identify the data source better than random guessing, and σ specifies the allowed deviation from this value. We fix $\sigma = 0.05$ across all tasks without tuning on downstream policy performance. Thus, t_{pm} is the earliest flow time at which the precision mismatch is no longer reliably detectable from x_t . We set $\mathcal{T}_{\text{pm}} = [t_{\text{pm}}, 1]$, retaining its target-aligned task context in the high-flow-time region while reducing the influence of source-dependent action differences in the noised input.

Precision-Matched, Task-Mismatched Boundary. The action-separability diagnostic used for $\mathcal{D}_{\text{pm}}$ is not suitable for $\mathcal{D}_{\text{tm}}$. Because different tasks naturally induce different action distributions, a source classifier may distinguish $\mathcal{D}_{\text{tm}}$ from $\mathcal{D}_{\text{pt}}$ even when their action precision is matched. The distinction should be reflected at the task context level rather than at the action level. We reuse action-context reliance derived from Section III-B to identify when task context becomes more influential than the noised action input.

At low flow times, prediction relies primarily on the noised action input, allowing $\mathcal{D}_{\text{tm}}$ to provide precise action supervision with limited influence from task mismatch. As flow time increases, context becomes more influential, making the task mismatch more consequential. Given the observed single crossover between the two reliance curves, we define t_{tm} as the latest flow time at which action reliance remains at least as strong as context reliance as shown in Figure 2 (right):

$$t_{\text{tm}} = \max \{t \in [0, 1] : d_{\text{act}}(t) \geq d_{\text{ctx}}(t)\} \quad (7)$$

We then set $\mathcal{T}_{\text{tm}} = [0, t_{\text{tm}}]$, retaining $\mathcal{D}_{\text{tm}}$ in the action-dominant regime while limiting the influence of its task-mismatched context.

D. Policy Integration

Given the computed admission boundaries, $\varepsilon 4\text{P}$ modifies policy training through source-dependent flow-time sampling. At each training step, we sample a source $s \sim \pi$ and an example $(a, c) \sim \mathcal{D}_s$, and then sample t uniformly from the corresponding admission region:

$$\mathcal{T}_s = \begin{cases} [0, 1], & s = \text{pt} \\ [t_{\text{pm}}, 1], & s = \text{pm} \\ [0, t_{\text{tm}}], & s = \text{tm} \end{cases} \quad (8)$$

The sampled t is used to construct x_t according to Equation (1), after which the policy is optimized with the standard flow-matching objective. Thus, $\mathcal{D}_{\text{pt}}$ supervises the full flow trajectory, while each imperfect source contributes only within its admitted region. The source sampling distribution π , model architecture, and training objective remain unchanged. The admission boundaries are computed once offline and remain fixed throughout training.

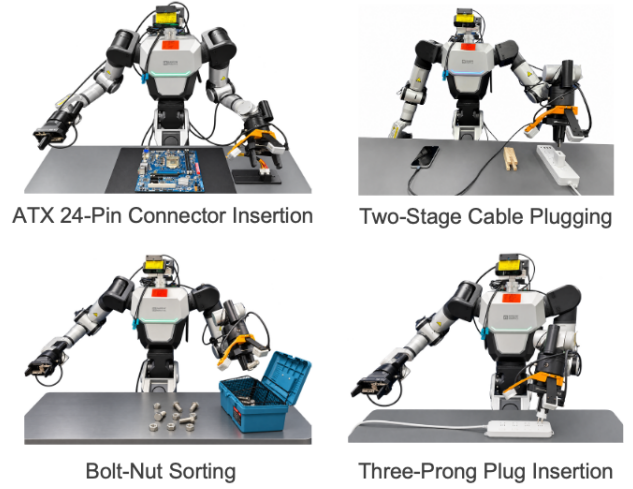


Fig. 4. Real-robot tasks used in the experiments. The target tasks are ATX 24-Pin Connector Insertion, Two-Stage Cable Plugging, and Bolt-Nut Sorting; Three-Prong Plug Insertion serves as the shared auxiliary dataset $\mathcal{D}_{\text{tm}}$ across three target tasks.

IV. EXPERIMENTAL SETUP

Hardware Setup. All experiments are conducted using the Rainbow Robotics RB-Y1 robot. High-precision demonstrations are collected through robot teleoperation, while lower-precision demonstrations are collected using our self-built UMI system. For both teleoperation and UMI data, the policy uses visual observations from two Intel RealSense D405 cameras. The UMI camera placement and gripper configuration are matched to those of the robot to minimize differences in viewpoint and embodiment.

Tasks and Data. All methods are evaluated on three real-robot manipulation tasks spanning different precision requirements: (1) ATX 24-Pin Connector Insertion requires inserting a 24-pin ATX connector into a motherboard header, with approximately 0.2 mm nominal clearance between each 3.3 mm pin and its 3.5 mm opening. (2) Two-Stage Cable Plugging requires first seating a two-prong plug and then inserting the USB-A end. The second stage requires approximately 0.3 mm nominal lateral clearance. (3) Bolt-Nut Sorting requires separating mixed bolts and nuts into their corresponding bins. This task has substantially larger placement tolerance and serves as a coarse-grained manipulation benchmark to assess whether the benefits of our method are specific to high-precision tasks. Together, these tasks evaluate performance across manipulation settings ranging from sub-millimeter-precision insertion to coarse object sorting. For brevity, we refer to the three tasks as ATX, Cable, and Sorting, respectively. Figure 4 gives an overview of the three target tasks.

For precision-mismatched data, we use UMI to collect $\mathcal{D}_{\text{pm}}$ for each target task. For task-mismatched data, we collect $\mathcal{D}_{\text{tm}}$ of inserting a three-prong plug, and use it as a shared auxiliary dataset across all three target tasks. We additionally collect $\mathcal{D}_{\text{pt}}^+$, a supplementary set of high-precision target-task demonstrations used to construct reference training sets for comparisons in which a portion of high-quality

TABLE I. Composition of the real-robot training datasets.

Dataset	Type	Demonstrations		
		ATX	Sorting	Cable
$\mathcal{D}_{\text{pt}}$	Teleop	300	100	100
$\mathcal{D}_{\text{pm}}$	UMI	100	50	50
$\mathcal{D}_{\text{tm}}$	Teleop	100	50	50
$\mathcal{D}_{\text{pt}}^+$	Teleop	100	50	50

TABLE II. Real-robot success rates (%) under native co-training, where each data source is admitted over the full flow-time axis $t \in [0, 1]$.

Training data	ATX	Sorting	Cable
$\mathcal{D}_{\text{pt}}$	48.3	81.7	78.3
$\mathcal{D}_{\text{pt}} + \mathcal{D}_{\text{pm}}$	33.3	75.0	25.0
$\mathcal{D}_{\text{pt}} + \mathcal{D}_{\text{tm}}$	43.3	78.3	71.7

data is replaced with imperfect data. Table I summarizes the collection source and number of demonstrations for each dataset and target task.

Evaluation. We adopt the pre-trained $\pi_{0.5}$ [3], [44] as the base policy, although $\epsilon 4\text{P}$ can be readily extended to other flow-matching policies. All evaluated methods use the same model architecture, pre-trained checkpoint, optimizer, batch size, source-sampling probabilities, and data and training budgets, and differ only in their source-dependent flow-time admission boundaries. Unless otherwise specified, we report the binary success rate over 60 real-robot trials per task.

V. EXPERIMENTS

We structure our experiments around four research questions to evaluate the effectiveness of the proposed method and understand how it leverages imperfect data beyond success rates:

- Under native co-training, does imperfect data help or harm policy learning?
- Can $\epsilon 4\text{P}$ effectively upcycle imperfect data to improve policy performance?
- Do the identified admission regions effectively capture where imperfect data is useful?
- Can $\epsilon 4\text{P}$ replace a portion of high-quality data with imperfect data?

A. Imperfect Data Is Harmful under Native Co-Training

Before evaluating the effectiveness of $\epsilon 4\text{P}$, we first empirically examine whether imperfect data provides useful or harmful supervision under native co-training, where each source is admitted over the full flow-time axis, i.e., $\mathcal{T}_s = [0, 1]$ for all $s \in \{\text{pt}, \text{pm}, \text{tm}\}$. As shown in Table II, training only on $\mathcal{D}_{\text{pt}}$ establishes baseline success rates of 48.3%, 81.7%, and 78.3% on ATX, Sorting, and Cable, respectively. Native co-training with $\mathcal{D}_{\text{pm}}$ substantially degrades performance on all three tasks, but the magnitude of

TABLE III. Real-robot success rates (%) with native co-training and $\epsilon 4\text{P}$. Bold and underlined values denote the best and second-best results, respectively, within each column.

Training data	Admission	ATX	Sorting	Cable	
				Stage 1	Stages 1–2
$\mathcal{D}_{\text{pt}}$	$\mathcal{T}_{\text{pt}}$	48.3	81.7	90.0	78.3
$\mathcal{D}_{\text{pt}} + \mathcal{D}_{\text{pm}}$	$[0, 1]$	33.3	75.0	70.0	25.0
$\mathcal{D}_{\text{pt}} + \mathcal{D}_{\text{tm}}$	$[0, 1]$	43.3	78.3	86.7	71.7
$\mathcal{D}_{\text{pt}} + \mathcal{D}_{\text{pm}} + \mathcal{D}_{\text{tm}}$	$[0, 1]$	38.3	73.3	75.0	33.3
$\mathcal{D}_{\text{pt}} + \mathcal{D}_{\text{pm}}$	$\mathcal{T}_{\text{pt}}, \mathcal{T}_{\text{pm}}$	<u>73.3</u>	93.3	<u>98.3</u>	90.0
$\mathcal{D}_{\text{pt}} + \mathcal{D}_{\text{tm}}$	$\mathcal{T}_{\text{pt}}, \mathcal{T}_{\text{tm}}$	71.7	88.3	100.0	81.7
$\mathcal{D}_{\text{pt}} + \mathcal{D}_{\text{pm}} + \mathcal{D}_{\text{tm}}$	$\mathcal{T}_{\text{pt}}, \mathcal{T}_{\text{pm}}, \mathcal{T}_{\text{tm}}$	80.0	<u>91.7</u>	<u>98.3</u>	<u>88.3</u>

degradation depends strongly on the precision requirement of the target task. Specifically, across the two high-precision tasks ATX and Cable, the success rate is reduced by 34.2 percentage points on average, compared with only a 6.7-percentage-point decrease on the coarse-grained Sorting task. This pattern suggests that lower-precision action supervision becomes increasingly harmful as successful task completion depends more strongly on fine-grained execution precision.

Co-training with $\mathcal{D}_{\text{tm}}$ exhibits a different failure pattern. Although it also reduces performance across all three tasks, the degradation is substantially smaller: 5.0 percentage points on ATX, 3.4 points on Sorting, and 6.6 points on Cable. Thus, $\mathcal{D}_{\text{tm}}$ introduces less severe degradation despite differing from the target tasks in context. This comparison suggests that different forms of mismatch affect native co-training in fundamentally different ways. Task-mismatched data behaves more like auxiliary multi-task supervision while preserving high-precision execution, whereas precision-mismatched data introduces imprecise action supervision directly on the target task, making it substantially more harmful for high-precision manipulation.

B. $\epsilon 4\text{P}$ Upcycles Imperfect Data to Improve Performance

Having shown that admitting imperfect data over the full flow-time axis causes source-dependent performance degradation, we next evaluate whether $\epsilon 4\text{P}$ can turn this otherwise harmful supervision into performance gains by restricting each source to its identified admission region. Table III summarizes the results.

For $\mathcal{D}_{\text{pm}}$, restricting admission to $\mathcal{T}_{\text{pm}}$ increases success on ATX from 33.3% to 73.3%, with substantial gains also observed on Sorting and the full Cable task. Similarly, restricting $\mathcal{D}_{\text{tm}}$ to $\mathcal{T}_{\text{tm}}$ increases success on ATX from 43.3% to 71.7% and consistently improves performance on the other tasks. The gain is particularly pronounced on the more challenging ATX task. Interestingly, $\mathcal{D}_{\text{pm}}$ is more harmful than $\mathcal{D}_{\text{tm}}$ under native co-training, yet performs better across all three tasks under $\epsilon 4\text{P}$. This reversal suggests that $\mathcal{D}_{\text{pm}}$ is highly useful but more sensitive to where it is admitted during training.

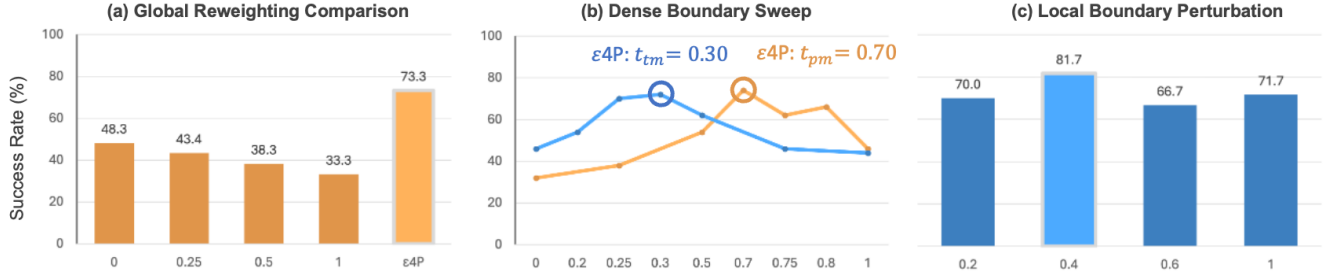


Fig. 5. (a) Global loss reweighting comparison using $\mathcal{D}_{pm}$ on ATX. (b) Dense boundary sweep for t_{pm} and t_{tm} on ATX. (c) Local boundary perturbation of t_{tm} on Cable.

TABLE IV. Real-robot success rates (%) on ATX under three admission strategies: native co-training, opposite-window control, and $\epsilon 4P$. Bold values denote the best result for each source.

Imperfect data	Method	Admission	Success
$\mathcal{D}_{pm}$	Native co-training	$[0, 1]$	33.3
	Opposite window	$[0, t_{pm}]$	30.0
	$\epsilon 4P$	$[t_{pm}, 1]$	73.3
$\mathcal{D}_{tm}$	Native co-training	$[0, 1]$	43.3
	Opposite window	$[t_{tm}, 1]$	46.7
	$\epsilon 4P$	$[0, t_{tm}]$	71.7

We then admit both imperfect sources within their respective admission regions. The resulting policy achieves 80.0% success on ATX, 91.7% on *Sorting*, and 88.3% full-task success on *Cable*, consistently outperforming both native co-training with the same sources and training with $\mathcal{D}_{pt}$ alone. The benefit of composition is clearest on ATX, where it outperforms either imperfect source alone. On *Sorting* and *Cable*, the best single-source policies already approach the performance ceiling, and composition therefore does not further improve their success rates. Nevertheless, it retains nearly all of their gains, suggesting that the two heterogeneous imperfect sources can occupy complementary flow-time regions within a single training run.

As an additional global loss reweighting baseline, we test whether simply reducing the influence of imperfect data can match the gains of source-dependent admission. On ATX, we fix the weight of $\mathcal{D}_{pt}$ to $w_{pt} = 1$ and keep $\mathcal{D}_{pm}$ active over the full flow-time range $[0, 1]$, while varying $w_{pm} \in \{0, 0.25, 0.5, 1\}$. Here, $w_{pm} = 0$ corresponds to using $\mathcal{D}_{pt}$ alone, whereas $w_{pm} = 1$ corresponds to native co-training. As shown in Figure 5(a), reducing w_{pm} mitigates the degradation caused by native co-training, but no nonzero weight outperforms the $\mathcal{D}_{pt}$ -only baseline. In contrast, $\epsilon 4P$ retains $w_{pm} = 1$ and restricts only its flow-time support, achieving the highest 73.3% success. This comparison shows that the benefit comes from controlling where, rather than merely how much, imperfect data contributes.

C. $\epsilon 4P$ Identifies Effective Admission Regions

We next examine whether the admission regions identified by $\epsilon 4P$ correspond to regions that are effective for policy

learning. We evaluate this through three complementary experiments: (1) a dense boundary sweep to assess global agreement, (2) a local perturbation of the identified t_{tm} to assess whether this boundary is locally preferred, and (3) opposite-window controls to test whether the identified regions reflect meaningful source-dependent structure rather than arbitrary design choices. All boundaries are computed using the offline diagnostics; the downstream evaluations are used only for validation.

First, we assess whether the admission boundaries identified by $\epsilon 4P$ align with those favored by downstream policy performance. On the most challenging ATX task, we compare the identified t_{pm} and t_{tm} against the performance landscape obtained from dense boundary sweeps. As shown in Figure 5(b), both boundaries lie within the high-performing regions. This agreement supports the effectiveness of the admission regions identified by $\epsilon 4P$.

Second, we assess whether an identified boundary t_{tm} is also locally preferred on a different task. Rather than performing another dense sweep, we evaluate the identified $t_{tm} = 0.4$ on the *Cable* task together with two neighboring values, $t_{tm} = 0.2$ and $t_{tm} = 0.6$. As shown in Figure 5(c), the identified boundary achieves the highest success among the evaluated values, providing additional local validation on a second task.

Finally, we test whether the selected admission regions are meaningful or whether any truncation of the flow-time range would provide similar benefits. As shown in Table IV, on ATX, $\mathcal{T}_{pm} = [t_{pm}, 1]$ achieves 73.3% success, compared with 30.0% using the opposite window $[0, t_{pm}]$ and 33.3% under native co-training. Similarly, $\mathcal{T}_{tm} = [0, t_{tm}]$ achieves 71.7%, compared with 46.7% using the opposite window $[t_{tm}, 1]$ and 43.3% under native co-training. These results show that performance depends on where each source is admitted, not merely on truncating its flow-time region.

Together, the global sweep, local perturbation, and opposite-window controls show that $\epsilon 4P$ identifies effective source-dependent admission regions that closely align with those favored by downstream policy performance.

D. $\epsilon 4P$ Reduces Reliance on High-Quality Data

Lastly, we examine whether $\epsilon 4P$ can reduce reliance on high-quality data by comparing source-dependent admission of imperfect data with training on additional perfect data $\mathcal{D}_{pt}^+$.

TABLE V. Real-robot success rates (%) comparing ε 4P with training on $\mathcal{D}_{\text{pt}}^+$. Bold and underlined values denote the best and second-best results, respectively, within each column.

Training data	Admission	ATX	Sorting	Cable	
				Stage 1	Stages 1–2
$\mathcal{D}_{\text{pt}}$	$\mathcal{T}_{\text{pt}}$	48.3	81.7	90.0	78.3
$\mathcal{D}_{\text{pt}} + \mathcal{D}_{\text{pt}}^+$	$\mathcal{T}_{\text{pt}}$	75.0	95.0	100.0	91.7
$\mathcal{D}_{\text{pt}} + \mathcal{D}_{\text{pm}}$	$\mathcal{T}_{\text{pt}}, \mathcal{T}_{\text{pm}}$	<u>73.3</u>	<u>93.3</u>	<u>98.3</u>	<u>90.0</u>
$\mathcal{D}_{\text{pt}} + \mathcal{D}_{\text{tm}}$	$\mathcal{T}_{\text{pt}}, \mathcal{T}_{\text{tm}}$	71.7	88.3	100.0	81.7

For a controlled comparison, $\mathcal{D}_{\text{pt}}^+$, $\mathcal{D}_{\text{pm}}$, and $\mathcal{D}_{\text{tm}}$ contain equal amounts of data.

As shown in Table V, replacing $\mathcal{D}_{\text{pt}}^+$ with $\mathcal{D}_{\text{pm}}$ yields closely comparable performance. Source-dependent admission of $\mathcal{D}_{\text{pm}}$ therefore recovers nearly all the benefit of augmenting $\mathcal{D}_{\text{pt}}$ with an equal amount of perfect data. The results with $\mathcal{D}_{\text{tm}}$ are less consistent, with gaps of 3.3 percentage points on ATX, 6.7 points on Sorting, and 10.0 points on the full Cable task. Thus, the extent to which imperfect data can replace high-quality data depends on the type and degree of mismatch. Overall, across both imperfect-data sources and all three tasks, ε 4P replaces an equal amount of additional high-quality data with suitably admitted imperfect data while incurring an average success-rate drop of only 4.2 percentage points.

VI. DISCUSSION AND FUTURE WORK

Discussion. Our results suggest that imperfect auxiliary data should not be accepted or rejected globally; its utility depends on where it contributes along the flow-time axis. $\mathcal{D}_{\text{pm}}$ and $\mathcal{D}_{\text{tm}}$ provide useful and harmful supervision at different flow times, with their effective admission regions lying at opposite ends of the axis. The opposite-window and loss reweighting controls further show that the gains depend on *where*, rather than simply *how much*, auxiliary supervision is applied.

The two admission boundaries capture distinct properties. The lower boundary t_{pm} is primarily data-dependent and identifies when source identity can no longer be reliably inferred from the noised action input. In contrast, the upper boundary t_{tm} is policy-dependent and reflects the relative sensitivity of the post-trained policy to its action and conditioning inputs. Neither boundary should be interpreted as an exact task-optimal threshold. Instead, they provide practical offline criteria for locating effective admission regions without exhaustive evaluation. The joint-training results further suggest that heterogeneous sources can coexist within one policy when assigned source-dependent admission regions.

Limitations and Future Work. Our experiments are currently limited to a single robot platform and a relatively small number of demonstrations collected in a controlled multi-camera setup. Although the experiments cover both high-precision and coarse-grained manipulation tasks, they do not yet fully evaluate the scalability and cross-embodiment generality of ε 4P. Future work will evaluate ε 4P across a

broader range of robot platforms and hardware configurations, and train policies with substantially larger and more diverse robot and human demonstration datasets.

VII. CONCLUSION

We present ε 4P, a source-dependent flow-time admission method for upcycling imperfect data in high-precision manipulation. Rather than uniformly co-training or globally reweighting imperfect sources, ε 4P admits precision-mismatched data at high flow times and task-mismatched data at low flow times, using boundaries derived from offline diagnostics. Real-robot experiments across three tasks show that this strategy turns otherwise harmful supervision into performance gains, improving success rates by up to 31.7 percentage points over training with perfect target-task data alone. Moreover, across both imperfect-data sources and all three tasks, ε 4P replaces an equal amount of additional perfect data with an average performance drop of only 4.2 percentage points. These findings establish flow time as a useful axis for integrating heterogeneous imperfect data, pointing toward a scalable paradigm for high-precision manipulation.

REFERENCES

- [1] M. J. Kim, K. Pertsch, S. Karamcheti, T. Xiao, A. Balakrishna, S. Nair, R. Rafailov, E. P. Foster, P. R. Sanketi, Q. Vuong, T. Kollar, B. Burchfiel, R. Tedrake, D. Sadigh, S. Levine, P. Liang, and C. Finn, “OpenVLA: An open-source vision-language-action model,” in *Proceedings of The 8th Conference on Robot Learning*, ser. Proceedings of Machine Learning Research, vol. 270. PMLR, 2025, pp. 2679–2713.
- [2] K. Black, N. Brown, D. Driess, A. Esmail, M. R. Equi, C. Finn, N. Fusai, L. Groom, K. Hausman, B. Ichter, S. Jakubczak, T. Jones, L. Ke, S. Levine, A. Li-Bell, M. Mothukuri, S. Nair, K. Pertsch, L. X. Shi, L. Smith, J. Tanner, Q. Vuong, A. Walling, H. Wang, and U. Zhilinsky, “ π_0 : A vision-language-action flow model for general robot control,” in *Proceedings of Robotics: Science and Systems*, Los Angeles, CA, USA, June 2025.
- [3] K. Black, N. Brown, J. Darpinian, K. Dhabalia, D. Driess, A. Esmail, M. R. Equi, C. Finn, N. Fusai, M. Y. Galliker, D. Ghosh, L. Groom, K. Hausman, B. Ichter, S. Jakubczak, T. Jones, L. Ke, D. LeBlanc, S. Levine, A. Li-Bell, M. Mothukuri, S. Nair, K. Pertsch, A. Z. Ren, L. X. Shi, L. Smith, J. T. Springenberg, K. Stachowicz, J. Tanner, Q. Vuong, H. Walke, A. Walling, H. Wang, L. Yu, and U. Zhilinsky, “ $\pi_{0.5}$: a vision-language-action model with open-world generalization,” in *Proceedings of The 9th Conference on Robot Learning*, ser. Proceedings of Machine Learning Research, vol. 305. PMLR, 2025, pp. 17–40.
- [4] J. Zhu, Z. Yan, Y. Xiao, J. Guo, C. Tie, X. Cao, Y. Zheng, and L. Shao, “ShapeForce: Low-cost soft robotic wrist for contact-rich manipulation,” in *IEEE International Conference on Robotics and Automation (ICRA)*, 2026.
- [5] A. Khazatsky, K. Pertsch, S. Nair, A. Balakrishna, S. Dasari, S. Karamcheti, S. Nasiriany, M. K. Srirama, L. Y. Chen, K. Ellis, P. D. Fagan, J. Hejna, M. Itkina, M. Lepert, Y. J. Ma, P. T. Miller, J. Wu, S. Belkale, S. Dass, H. Ha, A. Jain, A. Lee, Y. Lee, M. Memmel, S. Park, I. Radosavovic, K. Wang, A. Zhan, K. Black, C. Chi, K. B. Hatch, S. Lin, J. Lu, J. Mercat, A. Rehman, P. R. Sanketi, A. Sharma, C. Simpson, Q. Vuong, H. R. Walke, B. Wulfe, T. Xiao, J. H. Yang, A. Yavary, T. Z. Zhao, C. Agia, R. Bajjal, M. G. Castro, D. Chen, Q. Chen, T. Chung, J. Drake, E. P. Foster, J. Gao, D. A. Herrera, M. Heo, K. Hsu, J. Hu, D. Jackson, C. Le, Y. Li, R. Lin, Z. Ma, A. Maddukuri, S. Mirchandani, D. Morton, T. Nguyen, A. O’Neill, R. Scalise, D. Seale, V. Son, S. Tian, E. Tran, A. E. Wang, Y. Wu, A. Xie, J. Yang, P. Yin, Y. Zhang, O. Bastani, G. Berseth, J. Bohg, K. Goldberg, A. Gupta, A. Gupta, D. Jayaraman, J. J. Lim, J. Malik, R. Martín-Martín, S. Ramamoorthy, D. Sadigh, S. Song, J. Wu, M. C. Yip, Y. Zhu, T. Kollar, S. Levine, and C. Finn, “DROID: A large-scale

- in-the-wild robot manipulation dataset,” in *Proceedings of Robotics: Science and Systems*, Delft, Netherlands, July 2024.
- [6] Open X-Embodiment Collaboration, “Open X-Embodiment: Robotic learning datasets and RT-X models,” in *2024 IEEE International Conference on Robotics and Automation (ICRA)*, 2024, pp. 6892–6903.
 - [7] C. Tang, A. Xiao, Y. Deng, T. Hu, W. Dong, H. Zhang, D. Hsu, and H. Zhang, “MimicFunc: Imitating tool manipulation from a single human video via functional correspondence,” in *Proceedings of The 9th Conference on Robot Learning*, ser. Proceedings of Machine Learning Research, vol. 305. PMLR, 2025, pp. 4473–4492.
 - [8] H. Li, Y. Cui, and D. Sadigh, “How to train your robots? The impact of demonstration modality on imitation learning,” in *2025 IEEE International Conference on Robotics and Automation (ICRA)*, 2025, pp. 1113–1120.
 - [9] C. Chi, Z. Xu, C. Pan, E. Cousineau, B. Burchfiel, S. Feng, R. Tedrake, and S. Song, “Universal manipulation interface: In-the-wild robot teaching without in-the-wild robots,” in *Proceedings of Robotics: Science and Systems*, Delft, Netherlands, July 2024.
 - [10] S. Belkhal, Y. Cui, and D. Sadigh, “Data quality in imitation learning,” in *Advances in Neural Information Processing Systems*, vol. 36, 2023, pp. 80375–80395.
 - [11] Y. Mao, J. Fu, R. Zhang, H. Xie, and M. Yao, “Beyond success: Refining elegant robot manipulation from mixed-quality data via just-in-time intervention,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2026, pp. 13508–13518.
 - [12] J. Hejna, C. A. Bhateja, Y. Jiang, K. Pertsch, and D. Sadigh, “ReMix: Optimizing data mixtures for large scale imitation learning,” in *Proceedings of The 8th Conference on Robot Learning*, ser. Proceedings of Machine Learning Research, vol. 270. PMLR, 2025, pp. 145–164.
 - [13] Y.-H. Wu, N. Charoenphakdee, H. Bao, V. Tangkaratt, and M. Sugiyama, “Imitation learning from imperfect demonstration,” in *Proceedings of the 36th International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 97. PMLR, 2019, pp. 6818–6827.
 - [14] Y. Wang, C. Xu, B. Du, and H. Lee, “Learning to weight imperfect demonstrations,” in *Proceedings of the 38th International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 139. PMLR, 2021, pp. 10961–10970.
 - [15] H. Xu, X. Zhan, H. Yin, and H. Qin, “Discriminator-weighted offline imitation learning from suboptimal demonstrations,” in *Proceedings of the 39th International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 162. PMLR, 2022, pp. 24725–24742.
 - [16] A. Wei, N. Pfaff, T. Cohn, A. K. Dayı, C. Daskalakis, G. Daras, and R. Tedrake, “Ambient diffusion policy: Imitation learning from sub-optimal data in robotics,” in *Conference on Robot Learning (CoRL)*, 2026.
 - [17] Y. Lipman, R. T. Q. Chen, H. Ben-Hamu, M. Nickel, and M. Le, “Flow matching for generative modeling,” in *The Eleventh International Conference on Learning Representations*, 2023.
 - [18] D. Ghosh, H. R. Walke, K. Pertsch, K. Black, O. Mees, S. Dasari, J. Hejna, T. Kreiman, C. Xu, J. Luo, Y. L. Tan, L. Y. Chen, Q. Vuong, T. Xiao, P. R. Sanketi, D. Sadigh, C. Finn, and S. Levine, “Octo: An open-source generalist robot policy,” in *Proceedings of Robotics: Science and Systems*, Delft, Netherlands, July 2024.
 - [19] F. Lin, K. Arora, J. Mercat, H. Nishimura, P. Shah, C. Xu, M. Zhang, M. Zolotas, M. Angeles, O. Pfannenstiehl, A. Beaulieu, and J. Barreiros, “A systematic study of data modalities and strategies for co-training large behavior models for robot manipulation,” in *Proceedings of Robotics: Science and Systems*, Sydney, Australia, July 2026.
 - [20] L. Wang, J. Zhao, Y. Du, E. H. Adelson, and R. Tedrake, “PoCo: Policy composition from and for heterogeneous robot learning,” in *Proceedings of Robotics: Science and Systems*, Delft, Netherlands, July 2024.
 - [21] L. Wang, X. Chen, J. Zhao, and K. He, “Scaling proprioceptive-visual learning with heterogeneous pre-trained transformers,” in *Advances in Neural Information Processing Systems*, vol. 37, 2024.
 - [22] J. Hejna, S. Mirchandani, A. Balakrishna, A. Xie, A. Wahid, J. Thompson, P. R. Sanketi, D. Shah, C. M. Devin, and D. Sadigh, “Robot data curation with mutual information estimators,” in *Proceedings of Robotics: Science and Systems*, Los Angeles, CA, USA, June 2025.
 - [23] C. Agia, R. Sinha, J. Yang, R. Antonova, M. Pavone, H. Nishimura, M. Itkina, and J. Bohg, “CUPID: Curating data your robot loves with influence functions,” in *Proceedings of The 9th Conference on Robot Learning*, ser. Proceedings of Machine Learning Research, vol. 305. PMLR, 2025, pp. 2907–2932.
 - [24] K. Wu, N. Liu, Z. Zhao, D. Qiu, J. Li, Z. Che, Z. Xu, and J. Tang, “Learning from imperfect demonstrations with self-supervision for robotic manipulation,” in *2025 IEEE International Conference on Robotics and Automation (ICRA)*, 2025, pp. 16899–16906.
 - [25] A. Maddukuri, Z. Jiang, L. Y. Chen, S. Nasiriany, Y. Xie, Y. Fang, W. Huang, Z. Wang, Z. Xu, N. Chernyadev, S. Reed, K. Goldberg, A. Mandlikar, L. Fan, and Y. Zhu, “Sim-and-real co-training: A simple recipe for vision-based robotic manipulation,” in *Proceedings of Robotics: Science and Systems*, Los Angeles, CA, USA, June 2025.
 - [26] Y. Lei, M. Liu, A. Maddukuri, Z. Jiang, and Y. Zhu, “A mechanistic analysis of sim-and-real co-training in generative robot policies,” in *International Conference on Machine Learning (ICML)*, 2026.
 - [27] S. Kuhar, S. Cheng, S. Chopra, M. Bronars, and D. Xu, “Learning to discern: Imitating heterogeneous human demonstrations with preference and representation learning,” in *Proceedings of the 7th Conference on Robot Learning*, vol. 229, 2023, pp. 1437–1449.
 - [28] S. Yue, J. Liu, X. Hua, J. Ren, S. Lin, J. Zhang, and Y. Zhang, “How to leverage diverse demonstrations in offline imitation learning,” in *Proceedings of the 41st International Conference on Machine Learning*, vol. 235, 2024, pp. 58037–58067.
 - [29] J. Chen, H. Fang, H. Fang, and C. Lu, “Towards effective utilization of mixed-quality demonstrations in robotic manipulation via segment-level selection and optimization,” in *2025 IEEE International Conference on Robotics and Automation (ICRA)*, 2025, pp. 16884–16891.
 - [30] K. Gandhi, S. Karamcheti, M. Liao, and D. Sadigh, “Eliciting compatible demonstrations for multi-human imitation learning,” in *Proceedings of the 6th Conference on Robot Learning*, vol. 205, 2023, pp. 1981–1991.
 - [31] J. Ho, A. N. Jain, and P. Abbeel, “Denoising diffusion probabilistic models,” in *Advances in Neural Information Processing Systems*, vol. 33, 2020, pp. 6840–6851.
 - [32] B. Kawar, M. Elad, S. Ermon, and J. Song, “Denoising diffusion restoration models,” in *Advances in Neural Information Processing Systems*, vol. 35, 2022, pp. 23593–23606.
 - [33] Y. Zhu, K. Zhang, J. Liang, J. Cao, B. Wen, R. Timofte, and L. Van Gool, “Denoising diffusion models for plug-and-play image restoration,” in *2023 IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, 2023, pp. 1219–1229.
 - [34] Z. Wang, Z. Zhang, X. Zhang, H. Zheng, M. Zhou, Y. Zhang, and Y. Wang, “DR2: Diffusion-based robust degradation remover for blind face restoration,” in *2023 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2023, pp. 1704–1713.
 - [35] J. Choi, J. Lee, C. Shin, S. Kim, H. Kim, and S. Yoon, “Perception prioritized training of diffusion models,” in *2022 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022, pp. 11472–11481.
 - [36] T. Hang, S. Gu, C. Li, J. Bao, D. Chen, H. Hu, X. Geng, and B. Guo, “Efficient diffusion training via Min-SNR weighting strategy,” in *2023 IEEE/CVF International Conference on Computer Vision (ICCV)*, 2023, pp. 7441–7451.
 - [37] G. Daras, A. Rodriguez-Munoz, A. Klivans, A. Torralba, and C. Daskalakis, “Ambient diffusion omni: Training good models with bad data,” in *Advances in Neural Information Processing Systems*, vol. 38, 2025, pp. 42636–42673.
 - [38] C. Chi, Z. Xu, S. Feng, E. Cousineau, Y. Du, B. Burchfiel, R. Tedrake, and S. Song, “Diffusion policy: Visuomotor policy learning via action diffusion,” *The International Journal of Robotics Research*, vol. 44, no. 10–11, pp. 1684–1704, 2025.
 - [39] B. Chen, D. Martí Monsó, Y. Du, M. Simchowitz, R. Tedrake, and V. Sitzmann, “Diffusion forcing: Next-token prediction meets full-sequence diffusion,” in *Advances in Neural Information Processing Systems*, vol. 37, 2024, pp. 24081–24125.
 - [40] S. H. Høeg, Y. Du, and O. Egeland, “Fast policy synthesis with variable noise diffusion models,” in *2025 IEEE International Conference on Robotics and Automation (ICRA)*, 2025, pp. 4821–4828.
 - [41] M. M. Hong, J. Zhang, A. Nagabandi, and A. Gupta, “TMRL: Diffusion timestep-modulated pretraining enables exploration for efficient policy finetuning,” in *Proceedings of Robotics: Science and Systems*, Sydney, Australia, July 2026.
 - [42] D. Lopez-Paz and M. Oquab, “Revisiting classifier two-sample tests,” in *5th International Conference on Learning Representations (ICLR)*, 2017.

- [43] M. A. Pace, P. Dan, C. Ning, A. Bhardwaj, A. Du, E. W. Duan, W.-C. Ma, and K. Kedia, "X-Diffusion: Training diffusion policies on cross-embodiment human demonstrations," in *IEEE International Conference on Robotics and Automation (ICRA)*, 2026.
- [44] Physical Intelligence, "OpenPI: Open-source models and packages for robotics," [Online]. Available: <https://github.com/Physical-Intelligence/openpi>, 2025, accessed: Sep. 14, 2026.